

Notes: Chodorow-Reich & Falato (JF, 2022)

The Loan Covenant Channel: How Bank Health Transmits to the Real Economy

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1 Big picture

- **Question.** Why did lender health affect existing corporate borrowers during the 2008–2009 crisis even though most bank loans had long stated maturities?
- **Core answer.** Loan covenants created a channel through which unhealthy lenders could renegotiate or reduce long-term credit. A covenant breach puts the borrower into technical default and gives the lender bargaining power.
- **Main mechanism.** Bad lender health does not matter equally for every borrower. It matters especially when the borrower violates a covenant. The violation allows the lender to accelerate repayment, shrink the commitment, or force renegotiation.
- **Data contribution.** The paper uses supervisory Shared National Credit (SNC) data with loan-level covenant compliance, borrower identities, lender identities, loan amendments, and commitment changes.
- **Key fact 1: long maturity.** At the start of the crisis, most bank loans were not close to maturity. In Table I, roughly 88–89% of loans during 2008–2009 still had at least one year remaining.
- **Key fact 2: frequent violations.** Covenant breaches were common. In the lender-covenant sample, 33% of loans breached a covenant in a crisis year, and 37% breached in the current or previous year.
- **Main estimate.** In the baseline OLS specification, the interaction coefficient $Bad\ Lender \times Bind$ is 22.9 percentage points. Thus, moving from a healthier to a less healthy lender substantially increases the likelihood of credit reduction after a covenant breach.
- **Credit quantity effect.** In Table VI, unhealthy lenders reduce commitments especially for borrowers with covenant violations. The interaction effect is negative for total committed credit, drawn credit, and borrower-level aggregate credit.
- **Real effects.** In Table VII, borrowers squeezed by unhealthy lenders cut payouts, capital expenditures, and employment growth. They do not fully replace bank credit with non-SNC debt or bond issuance.
- **Robustness.** The result survives alternative lender-health measures, sample restrictions, placebo tests, and dropping one lender at a time.
- **Aggregation.** The covenant channel accounts for a large share of the cross-sectional variation in bank credit supply during the crisis: about 4.9% of total credit in 2008 and 5.2% in 2009, compared with bank-level contractions of 8.6% and 6.1%.
- **Economic takeaway.** Long-term loan contracts are not fully insulated from bank distress. Covenants create state-contingent control rights that can transmit shocks from lender balance sheets to borrower credit and real activity.

2 Data and measurement

2.1 SNC loans and the covenant compliance sample

- The main data come from the U.S. Shared National Credit supervisory data set.
- The SNC universe covers large syndicated loans and tracks commitments, outstanding balances, borrower identities, lead lenders, and loan characteristics.
- The authors add a supervisory covenant compliance sample that records whether a loan breaches a covenant and how lenders respond.
- The full data cover roughly \$2 trillion of loan commitments at the start of 2008.
- The paper focuses on nonfinancial borrowers with credit lines or term loans.

Interpretation. The paper’s empirical advantage is that it observes covenant compliance directly. This matters because public filings miss many violations that are waived or amended before reporting dates.

2.2 Loan maturity and covenant violations

- Stated loan maturity is long: in the crisis period, the mean maturity remaining for loans with more than one year left is about 2.6–2.8 years.
- Long maturity alone would suggest limited immediate exposure to bank distress.
- But covenant violations are common: during 2008–2009, 34% of loans in the covenant sample and 33% in the lender-covenant sample breached a covenant in a given year.
- The authors use

$$Bind_{l,t-1:t} = \max\{Bind_{l,t-1}, Bind_{l,t}\},$$

which equals one if loan l breaches a covenant in the current or previous year.

Interpretation. The timing recognizes that lender-borrower renegotiation may take months. A violation in the previous year can still affect credit outcomes in the current year.

2.3 Main outcome variables

- *Credit Reduced* equals one if the loan is terminated before maturity or the loan commitment is reduced.
- $\% \Delta Total\ Committed$ measures the percent change in total loan commitment.
- Additional outcomes include loan termination, intensive-margin commitment reduction, utilization changes, drawn credit, non-SNC debt, bond issuance, payouts, capital expenditures, and employment growth.

Interpretation. The paper distinguishes the extensive margin of a credit reduction from the intensive margin of how much credit is cut.

2.4 Lender health

- Lender health is measured using three preexisting crisis-exposure measures from Chodorow-Reich (2014):
 - exposure to Lehman Brothers in syndication portfolios;
 - correlation with the ABX AAA 2006-H1 mortgage-backed securities index;
 - 2007–2008 trading revenue as a share of assets.

- The paper extracts the first principal component and rank-normalizes it to form $Bad\ LenderLender_b$.
- $Bad\ LenderLender_b \in [0, 1]$, with one corresponding to the least healthy lender.
- In the main results, lender health is assigned using the lead lender.

Interpretation. The lead lender matters because it organizes the syndicate, manages servicing, and typically plays a key role in renegotiation after a covenant breach.

2.5 Borrower and loan controls

The main specifications include borrower and loan controls such as:

- borrower log assets, leverage, and risk rating;
- loan purpose and loan type;
- year and industry fixed effects;
- interactions of borrower quality controls with lender health.

Interpretation. These controls are intended to isolate the effect of lender health conditional on borrower quality, and to reduce concern that unhealthy lenders simply had worse borrowers.

3 Models and empirical specifications

3.1 Core loan covenant channel

The mechanism can be summarized as

bad lender health $\Rightarrow$ higher shadow cost of funds $\Rightarrow$ desire to reduce credit

covenant violation $\Rightarrow$ technical default and renegotiation rights $\Rightarrow$ credit reduction possible even before maturity.

Interpretation. The key interaction is between lender distress and borrower technical default. Bad lender health alone does not necessarily give the lender the contractual right to reduce long-term credit; covenant violation supplies that right.

3.2 Baseline regression

The main specification is

$$Y_{l,b,f,t} = \beta_0 + \beta_1[Bad\ Lender_b] + \beta_2[Bind_{l,t-1:t}] + \beta_3[Bad\ Lender_b \times Bind_{l,t-1:t}] + \gamma' X_{l,b,f,t} + \varepsilon_{l,b,f,t}, \quad (1)$$

where:

- l indexes loans, b lenders, f firms, and $t \in \{2008, 2009\}$;
- $Y_{l,b,f,t}$ is a credit or firm outcome;
- $Bad\ Lender_b$ is lender health, with larger values meaning worse health;
- $Bind_{l,t-1:t}$ is covenant violation in the current or previous year;
- β_3 is the main coefficient of interest.

Interpretation. β_3 measures whether covenant violators are treated more harshly when their lead lender is less healthy.

3.3 Near-threshold design

The near-threshold sample keeps loans close to a covenant threshold. The logic is regression-discontinuity style:

- borrowers just below the threshold violate;
- borrowers just above the threshold do not;
- if there is no precise manipulation, violation status is locally as good as random.

Interpretation. This design reduces concern that covenant violators are fundamentally worse borrowers than nonviolators. The paper reports McCrary tests that do not reject continuity of the distance-to-threshold density.

3.4 Instrumental variables design

The IV design treats $Bind$ and $Bad\ Lender \times Lender$ as endogenous. The excluded instruments are:

- an indicator for whether the bank had a strict or lax supervisor at loan origination;
- that supervisor-strictness indicator interacted with $Bad\ Lender \times Lender$.

The first stage relies on stricter supervisors inducing tighter covenants at origination and thus making violations more likely later.

Interpretation. The exclusion restriction is that supervisor strictness at origination affects crisis-period credit outcomes only through the probability of covenant violation, conditional on crisis-period supervisor strictness and controls.

3.5 Heterogeneity specification

For heterogeneity tests, the paper estimates

$$Y_{l,b,f,t} = \beta_0 + \beta_1[Bad\ Lender_b] + \beta_2[Bind] + \beta_3[Bad\ Lender_b \times Bind] + \gamma'X_{l,b,t} \quad (2)$$

$$+ \beta_{0,I}I + \beta_{1,I}[Bad\ Lender_b \times I] + \beta_{2,I}[Bind \times I] + \beta_{3,I}[Bad\ Lender_b \times Bind \times I] + \gamma'_I[X_{l,b,t} \times I] + \varepsilon_{l,b,f,t}. \quad (3)$$

Interpretation. The coefficient β_3 gives the treatment effect when $I = 0$. The sum $\beta_3 + \beta_{3,I}$ gives the treatment effect when $I = 1$.

3.6 Aggregation formula

The fitted decline in credit due to the covenant channel is

$$\text{Fitted decline} = -(0.01 \times \beta_3) \sum_l Bad\ Lender_b \times Bind_{l,t-1:t} \times Commit_{l,t-1}. \quad (4)$$

Dividing this fitted decline by beginning-of-year committed credit gives the aggregate share of credit lost because unhealthy lenders squeezed covenant violators.

Interpretation. This exercise asks how much of the actual cross-sectional lender credit contraction can be accounted for by the covenant channel, not what would have happened in a counterfactual world with no covenants.

4 Identification and empirical design

4.1 Threat 1: unhealthy lenders may have worse borrowers

- Borrowers of unhealthy lenders could be riskier before the crisis.
- Table II addresses this by comparing precrisis borrower and loan characteristics across healthier and less healthy lenders.
- The authors show balance in log assets, leverage, risk rating, maturity remaining, cash-flow variables, volatility, and violation propensity.

Interpretation. Balance does not prove random assignment, but it weakens the concern that the main effect is driven by ex ante borrower differences.

4.2 Threat 2: covenant violators are worse borrowers

- Covenant violators are likely weaker than nonviolators.
- The baseline specification controls for borrower quality and interacts borrower quality with lender health.
- The near-threshold design compares borrowers close to a covenant threshold.
- The IV design uses supervisor strictness at origination as variation in the probability of violation.

Interpretation. The strongest causal interpretation rests on the idea that, near the threshold or under the IV, violation status is not simply a proxy for unobserved borrower distress.

4.3 Threat 3: bank health may be endogenous to corporate lending

- The crisis shock was primarily driven by real estate, MBS, trading, and counterparty exposures rather than nonfinancial corporate borrowers.
- The lender-health measure uses exposure to the financial crisis, not contemporaneous loan performance in the sample.
- Figure 1 shows that healthier and less healthy lenders had stable shares before the crisis, then diverged in 2008.

Interpretation. The lender-health variation is plausibly supply-side because the sources of bank distress are external to most nonfinancial borrowers.

4.4 Threat 4: a single lender or sample definition drives the result

- Table VIII uses alternative lender-health measures.
- Table IX changes the sample, reweights risk ratings, drops existing violators, restricts to two-year maturity loans, and adds alternative covenant data sources.
- Figure 3 drops one lender at a time.
- Table X conducts placebo tests.

Interpretation. The main interaction survives many robustness checks, which supports the interpretation that the loan covenant channel is systematic rather than idiosyncratic.

4.5 What the paper cannot fully prove

- Covenant violation is not randomly assigned in the full OLS sample.
- Borrowers could manipulate covenant thresholds, although density tests do not show clear evidence of this.
- The IV exclusion restriction is not directly testable.
- The aggregation exercise is not a no-covenants counterfactual; banks might have reduced credit on other margins if covenants were unavailable.

Interpretation. The paper's strength is not a single perfect experiment but the convergence of evidence from administrative data, balancing tests, near-threshold comparisons, IV, robustness checks, and aggregation.

5 Tables and figures

This section follows the same compact *Read* plus *Economic message* format. Original figures and tables are directly cropped from the paper and grouped where possible.

5.1 Table I and Figure 1: long maturity, frequent violations, and lender health

Read:

- Table I shows that most loans are long-term: in 2008–2009, 88–89% of loans have at least one year remaining.
- Mean maturity remaining for long-term loans is about 2.6–2.8 years during the crisis.
- The share with *Credit Reduced* = 1 rises from 26% precrisis to 38% in the lender-covenant sample during the crisis.
- Covenant violations are common: $Bind_t$ equals 0.33 and $Bind_{t-1:t}$ equals 0.37 in the crisis lender-covenant sample.
- Figure 1 shows that least healthy lenders' aggregate commitment share drops sharply in 2008, while healthiest lenders' share rises.

Economic message. Long maturity does not prevent credit contraction because covenants create renegotiation rights before maturity.

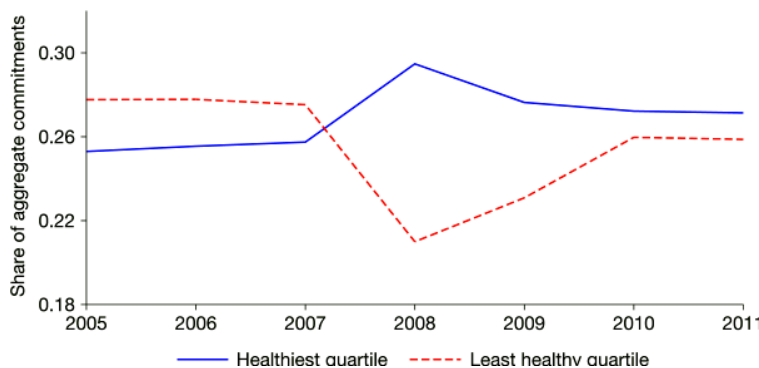


Figure 1: Figure 1: Aggregate lending by quartile of bank health.

Table I
Summary Statistics

The table reports summary statistics for the precrisis (2006 to 2007) and crisis (2008 to 2009) periods and for three samples. Columns with the header “Universe” report summary statistics for the universe of credit lines and term loans to nonfinancial borrowers in the full SNC data set. Columns with the header “Covenant” report summary statistics for the subset of these loans in the SNC covenant sample. Columns with the header “Lender-covenant” report summary statistics for our final sample of all credit lines and term loans in the covenant sample to nonfinancial borrowers and for which the lead lender is in the Chodorow-Reich (2014) lender health data set. *Credit Reduced* equals 1 if either the loan is terminated before maturity or the loan commitment is reduced. $Bind_t$ and $Bind_{t-1,t}$ are indicator variables equal to 1 if a loan breached a covenant in the current year or in either the current or previous year, respectively. Total committed is the sum of loans outstanding and unused commitments averaged over the two-year period.

Sample:	Precrisis (2006–2007)			Crisis (2008–2009)		
	Universe (1)	Cove- nant (2)	Lender- Cove- nant (3)	Universe (4)	Cove- nant (5)	Lender- Cove- nant (6)
<i>Loans of Any Maturity</i>						
Share 1+ year remaining	0.91	0.92	0.92	0.88	0.88	0.89
<i>Loans with 1+ Year Maturity Remaining</i>						
<i>Loan characteristics</i>						
Mean maturity (years)	3.30	3.34	3.36	2.61	2.78	2.75
2+ years (share)	0.85	0.88	0.89	0.74	0.79	0.79
Log committed (mean)	18.66	18.85	18.95	18.74	18.76	18.91
Credit line (share)	0.61	0.59	0.60	0.51	0.49	0.49
<i>Credit Reduced</i> (share)	0.27	0.26	0.26	0.37	0.38	0.38
Lead share (mean)	0.19	0.14	0.15	0.18	0.14	0.14
Utilization rate (mean)	0.53	0.56	0.55	0.61	0.64	0.63
<i>Borrower characteristics</i>						
Publicly-traded (share)	0.37	0.38	0.40	0.36	0.37	0.39
Log assets (mean)		12.58	12.76		12.68	12.82
Leverage (mean)		0.49	0.49		0.53	0.50
Passing risk rating (%)	88.79	70.78	70.68	75.53	47.17	45.96
Or special mention (%)	95.92	89.32	88.92	85.42	71.31	70.94
<i>Covenant violations</i>						
$Bind_t$		0.25	0.24		0.34	0.33
$Bind_t$, private borrowers		0.27	0.27		0.36	0.34
$Bind_t$, excluding waivers		0.09	0.08		0.11	0.10
$Bind_{t-1,t}$		0.29	0.28		0.39	0.37
Loan-year observations	11,247	2,676	2,478	11,979	4,059	3,420
Unique borrowers	4,769	1,309	1,166	4,992	1,704	1,409
Total committed (\$Tr)	2.01	0.55	0.50	2.04	0.72	0.65

Figure 2: Table I: Summary statistics.

5.2 Table II and Table III: balance and nonparametric evidence

Read:

- Table II shows that borrowers of less healthy and more healthy lenders look similar on precrisis characteristics.
- Among violators, log assets, leverage, risk rating, maturity, cash flow, ROA, Tobin’s Q, volatility, and other variables are balanced.
- Table III gives nonparametric evidence: for healthiest lenders, violation raises the credit-reduction probability by 5.3 percentage points.
- For least healthy lenders, violation raises that probability by 18.6 percentage points.
- The difference-in-differences is 13.3 percentage points for *Credit Reduced* and -13.1 percentage points for percent change in total committed credit.

Economic message. The raw data already show the central pattern: covenant violations matter much more when the lender is unhealthy.

Table II
Balancing

The table reports variable means by lender health. “Less healthy” lenders are those for which *Bad Lender* > median and “More healthy” lenders are those for which *Bad Lender* < median, where *Bad Lender* is the rank of the lead lender’s health normalized to lie on the unit interval, with a value of 1 corresponding to the least healthy lender.

Lender Health:	All Borrowers			$Bind_{t-1,t} = 1$		
	Less Healthy (1)	More Healthy (2)	<i>t</i> -Stat. Equality (3)	Less Healthy (4)	More Healthy (5)	<i>t</i> -Stat. Equality (6)
<i>Panel A: SNC Precrisis Variables</i>						
Covenant tightness	1.72	5.74	0.23			
Log assets	12.72	12.81	1.17	10.98	11.11	0.35
Leverage	0.50	0.49	1.21	0.54	0.53	0.93
Risk rating	70.04	71.51	0.56	42.20	44.69	0.93
Maturity remaining	3.34	3.39	1.12	2.66	2.69	0.36
Observations	1,215	1,263	2,478	358	335	693
<i>Panel B: Compustat Precrisis Variables</i>						
Cash flow/assets (%)	4.86	4.89	0.04	0.58	0.41	0.08
ROA (%)	10.22	10.49	0.30	8.66	9.41	0.63
Tobin's Q	1.52	1.43	0.90	1.36	1.29	0.45
Z-score	0.16	0.09	1.09	0.08	0.06	0.52
Disc. accruals/assets (%)	-0.35	-1.16	0.53	-1.66	-1.75	0.36
S&P credit rating (%)	76.75	74.60	0.63	76.56	78.78	0.30
Bankruptcy or delisting	5.67	6.85	1.18	7.81	8.09	0.26
Market beta	3.65	3.77	-0.12	3.46	3.39	0.07
Idiosyncratic volatility	12.62	12.76	0.46	13.03	12.63	0.76
Total volatility	21.99	21.77	1.02	22.25	22.21	0.11
Observations	450	469	919	124	133	257
<i>Panel C: SNC Crisis Variables</i>						
$100 \times Bind_{t-1,t}$	37.96	36.59	0.82			
Observations	1,673	1,747	3,420			

(a) Table II: Balancing

Table III
Nonparametric Evidence

The table reports the fraction of loans with a reduction in commitment (Panel A) or the average change in loan commitment (Panel B) in each cell. The sample consists of all loans in the SNC covenant sample at the start of 2008 or 2009, with at least one year maturity remaining, and with a lead lender in the lender health data set. *Bad Lender* is the rank of the lead lender’s health normalized to lie on the unit interval, with a value of 1 corresponding to the least healthy lender. *Bind* is an indicator variable equal to 1 if a borrower violated a covenant in the current or the previous year. The brackets report the number of observations in each cell.

Lender Health Quartile	$Bind_{t-1,t} = 0$	$Bind_{t-1,t} = 1$	Difference
<i>Panel A: Fraction Credit Reduced = 1</i>			
Quartile 1 (healthiest lenders)	0.316 [N = 529]	0.369 [N = 319]	0.053
Quartile 2	0.329 [N = 517]	0.405 [N = 343]	0.077
Quartile 3	0.282 [N = 507]	0.435 [N = 351]	0.153
Quartile 4 (least healthy lenders)	0.320 [N = 489]	0.506 [N = 365]	0.186
Difference Q4-Q1	0.004	0.137	0.133
<i>Panel B: %Δ Total Committed</i>			
Quartile 1 (healthiest lenders)	0.004 [N = 529]	-0.065 [N = 319]	-0.069
Quartile 2	0.024 [N = 517]	-0.074 [N = 343]	-0.098
Quartile 3	0.039 [N = 507]	-0.131 [N = 351]	-0.172
Quartile 4 (least healthy lenders)	0.044 [N = 489]	-0.156 [N = 365]	-0.200
Difference Q4-Q1	0.040	-0.091	-0.131

(b) Table III: Nonparametric evidence

5.3 Tables IV–V: baseline OLS, near-threshold, and IV estimates

Read:

- Table IV column (3), the baseline controlled specification, estimates $Bad\ Lender \times Lender \times Bind = 22.9$ percentage points for *Credit Reduced*.
- Column (4), including expiring loans, gives 17.6 percentage points.
- Column (5) shows the interaction is 10.9 percentage points for outright termination.
- Column (6) shows a 12.0 percentage point intensive-margin reduction.
- Table V confirms the result in alternative designs: 23.0 percentage points in the near-threshold sample and 24.4 percentage points in IV.

Economic message. The main coefficient is large and robust across identification strategies. Bad lender health turns covenant violations into credit reductions.

Table IV
OLS Estimates of Loan Terminated or Reduced

The table reports linear probability model regressions of the form $Y_{i,b,t} = \beta_0 + \beta_1[\text{Bad Lender}] + \beta_2[\text{Bind}] + \beta_3[\text{Bad Lender} \times \text{Bind}] + \gamma'X_{i,b,t} + \epsilon_{i,b,t}$. The sample consists of all loans in the SNC covenant sample at the start of 2008 or 2009 and a lead lender in the lender health data set. The variable *Credit Reduced* equals 1 if the loan is terminated before maturity (*Term.*) or the loan commitment is reduced (*Intensive margin*). *Bad Lender* is the rank of the lead lender's health normalized to lie on the unit interval, with a value of 1 corresponding to the least healthy lender. *Bind* is an indicator variable equal to 1 if a borrower violated a covenant in either the current or previous year. Reported coefficients are multiplied by 100. Borrower controls: log assets, leverage, risk rating. Loan controls: loan purpose, loan type. Standard errors two-way clustered by borrower and lead lender are reported in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively.

Dependent Variable:	Credit Reduced				Term.	Intensive Margin
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Bad Lender</i>	8.8** (4.4)	-3.8 (5.9)	5.8 (12.1)	0.5 (10.3)	1.5 (5.0)	4.3 (13.6)
<i>Bind</i>	14.1*** (2.3)	4.3 (3.1)	5.6** (2.7)	6.4** (2.6)	0.7 (3.3)	4.9 (3.1)
<i>Bad Lender</i> $\times$ <i>Bind</i>		25.9*** (6.7)	22.9*** (6.7)	17.6** (6.7)	10.9** (5.4)	12.0** (5.2)
Estimator	OLS	OLS	OLS	OLS	OLS	OLS
Year, Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Borrower, loan controls	No	No	Yes	Yes	Yes	Yes
Controls $\times$ <i>Bad Lender</i>	No	No	Yes	Yes	Yes	Yes
Include expiring loans	No	No	No	Yes	No	No
R^2	0.080	0.081	0.116	0.114	0.107	0.100
Observations	3,420	3,420	3,420	3,843	3,420	3,420

(a) Table IV: OLS estimates

Table V
Alternative Estimates of Loan Terminated or Reduced

The table reports linear probability model regressions of the form $Y_{i,b,t} = \beta_0 + \beta_1[\text{Bad Lender}] + \beta_2[\text{Bind}] + \beta_3[\text{Bad Lender} \times \text{Bind}] + \gamma'X_{i,b,t} + \epsilon_{i,b,t}$. The sample consists of all loans in the SNC covenant sample at the start of 2008 or 2009 with at least one year maturity remaining and a lead lender in the lender health data set. The column (1) sample additionally restricts to loans with distance to the closest covenant threshold of less than 30%. The variable *Credit Reduced* equals 1 if either the loan is terminated before maturity or the loan commitment is reduced. *Bad Lender* is the rank of the lead lender's health normalized to lie on the unit interval, with a value of 1 corresponding to the least healthy lender. *Bind* is an indicator variable equal to 1 if a borrower violated a covenant in either the current or previous year. Reported coefficients are multiplied by 100. Borrower controls: log assets, leverage, risk rating. Loan controls: loan purpose, loan type. Standard errors are two-way clustered by borrower and lead lender reported in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively.

Dependent Variable:	Credit Reduced	
	(1)	(2)
<i>Bad Lender</i>	11.1 (10.4)	4.4 (3.9)
<i>Bind</i>	6.7 (4.3)	6.5 (14.7)
<i>Bad Lender</i> $\times$ <i>Bind</i>	23.0** (11.0)	24.4*** (6.8)
Estimator	OLS	IV
Close violator sample	Yes	No
Year, Industry FE	Yes	Yes
Borrower, loan controls	Yes	Yes
Distance control	Yes	No
2008-09 supervisor control	No	Yes
Controls $\times$ <i>Bad Lender</i>	Yes	Yes
Observations	780	3,420

(b) Table V: Alternative estimates

5.4 Table VI and Figure 2: total credit around the covenant threshold

Read:

- Table VI examines credit quantities rather than just whether a reduction occurs.
- In the full-sample OLS panel, *Bad Lender* $\times$ *Bind* equals -16.4 percentage points for loan intensive margin, -24.4 for all loan margins, and -30.1 at the borrower level.
- The near-threshold OLS estimates are even larger: -24.6, -34.8, and -35.1 percentage points.
- IV estimates remain negative for total committed credit and drawn credit.
- Figure 2 shows the discontinuity visually: loans in violation with less healthy lenders experience a sharp decline in committed credit.

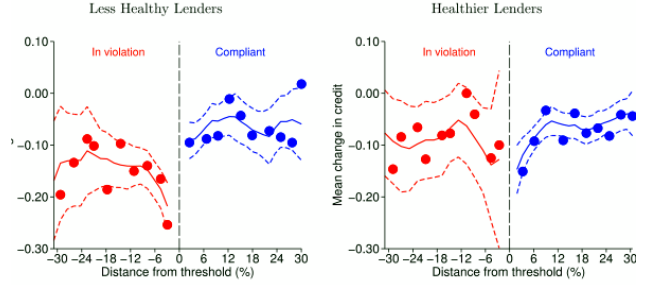
Economic message. The covenant channel affects actual credit quantities, not merely formal contract status.

Table VI
Effect on Total Credit

The table reports regressions of the form $Y = \beta_0 + \beta_1[Bad\ Lender] + \beta_2[Bind] + \beta_3[Bad\ Lender \times Bind] + \gamma'X + \epsilon$. *Bad Lender* is the rank of the lead lender's health normalized to lie on the unit interval, with a value of 1 corresponding to the least healthy lender. *Bind* is an indicator variable equal to 1 if a borrower violated a covenant in the current or previous year. The column (1) sample excludes loans that disappear or change identifier by the end of the year; the column (2) sample is the same as Table IV; in columns (3) to (5), the sample contains all loans in the SNC universe to a borrower in the Table IV sample and the data are collapsed to the borrower-year level. The dependent variable is the percent change in total committed credit associated with loan l (1); the percent change in total committed credit on loans from lead lender b to borrower f (2); the percent change in total committed credit aggregated across all loans in the SNC universe to borrower f (3); the change in the utilization rate across all loans to borrower f (4); or the percent change in total credit outstanding (the sum of term loans and the drawn portion of credit lines) aggregated across all loans to borrower f in the SNC universe (5). SNC borrower controls: log assets, leverage, risk rating. Loan controls: loan purpose, loan type. Standard errors two-way clustered by borrower and lead lender (columns (1) and (2)) or by borrower and worst lead lender (columns (3) to (5)) reported in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively.

Dependent Variable:	%Δ Total Committed			Δ Utilization	%Δ Drawn
Aggregation:	Loan intensive margin (1)	Loan all margins (2)	Borrower (3)	Borrower (4)	Borrower (5)
Panel A: Full Sample OLS					
<i>Bad Lender</i>	-0.9 (3.3)	6.3 (14.0)	6.2 (13.4)	0.3 (5.7)	7.4 (6.9)
<i>Bind</i>	-1.0 (1.3)	-8.5** (3.7)	-6.3** (3.1)	-1.5 (2.6)	2.3 (3.1)
<i>Bad Lender × Bind</i>	-16.4*** (3.3)	-24.4*** (14.0)	-30.1*** (10.3)	7.3*** (1.5)	-19.1** (8.0)
Panel B: Near Violation Threshold OLS					
<i>Bad Lender</i>	5.4 (10.0)	1.8 (12.5)	-1.9 (18.7)	3.7 (4.8)	8.0 (10.1)
<i>Bind</i>	-5.1 (4.5)	-10.3 (7.1)	-7.8 (7.6)	-2.8 (8.4)	8.6 (5.2)
<i>Bad Lender × Bind</i>	-24.6*** (8.3)	-34.8*** (11.3)	-35.1** (17.7)	10.3** (4.5)	-34.8** (14.5)
Panel C: Full Sample IV					
<i>Bad Lender</i>	1.3 (3.5)	0.8 (17.5)	8.5 (13.9)	2.3 (2.0)	4.4 (9.1)
<i>Bind</i>	-1.6 (3.8)	-9.6 (7.6)	-5.9 (5.1)	-1.2 (3.4)	3.6 (5.9)
<i>Bad Lender × Bind</i>	-18.9*** (6.2)	-30.6*** (8.4)	-34.5*** (7.5)	9.6** (3.8)	-23.7*** (6.0)
Year, Industry FE	Yes	Yes	Yes	Yes	Yes
Borrower controls	Yes	Yes	Yes	Yes	Yes
Loan controls	Yes	Yes	No	No	No
Distance control	Panel B	Panel B	Panel B	Panel B	Panel B
2008-09 supervisor control	Panel C	Panel C	Panel C	Panel C	Panel C
Controls × <i>Bad Lender</i>	Yes	Yes	Yes	Yes	Yes
Panels A&C observations	2,289	3,420	1,803	1,803	1,803

(a) Table VI: Effect on total credit



(b) Figure 2: Committed credit around covenant threshold

5.5 Table VII: financial and real adjustment

Read:

- Borrowers do not fully replace lost bank credit with non-SNC debt or bond issuance.
- In the full-sample OLS panel, *Bad Lender* × *Bind* is -4.5 for payouts/assets, -2.7 for capex/assets, and -15.6 for employment growth.
- In the full-sample IV panel, the corresponding effects are -8.4, -4.9, and -16.9.
- The estimates are strongest for payout, capital expenditure, and employment outcomes.

Economic message. The covenant channel transmits from bank health to the real economy: credit cuts lead firms to reduce investment and employment.

Table VII
Financial and Real Adjustment

The table reports regressions of the form $Y = \beta_0 + \beta_1[Bad\ Lender] + \beta_2Bind + \beta_3[Bad\ Lender \times Bind] + \gamma'X + \epsilon$. The column (1) sample contains all borrowers in the Table IV sample with at least one SNC loan outstanding at the end of the year and the data are collapsed to the borrower-year level. In columns (2) to (5), the sample comprises borrowers in the Table IV sample that also appear in Compustat and the data are collapsed to the borrower-year level. *Bad Lender* is the rank of the lead lender's health normalized to lie on the unit interval, with a value of 1 corresponding to the least healthy lender. *Bind* is an indicator variable equal to 1 if a borrower violated a covenant in either the current or previous year. The dependent variable is the change in non-SNC debt as a share of assets (1); bond issuance as a share of assets (2); shareholder payouts as a share of assets (3); capital expenditure as a share of assets (4); or the log change in employment (5), winsorized at the 1% level in columns (2) to (5). Borrower controls: log assets, leverage, risk rating. Standard errors are two-way clustered by borrower and worst lead lender reported in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively.

Dependent Variable:	$\Delta Non-SNC$ <i>Debt/Assets</i> (1)	<i>Bond</i> <i>Issues/</i> <i>Assets</i> (2)	<i>Payouts/</i> <i>Assets</i> (3)	<i>Capex/</i> <i>Assets</i> (4)	<i>Emp.</i> <i>Growth</i> (5)
<i>Panel A: Full Sample OLS</i>					
<i>Bad Lender</i>	9.0 (13.8)	-1.7 (2.7)	-5.6 (6.1)	5.2 (3.4)	-2.4 (5.4)
<i>Bind</i>	6.6 (6.2)	-0.9 (1.2)	0.9 (1.6)	0.2 (0.4)	-0.2 (3.1)
<i>Bad Lender</i> $\times$ <i>Bind</i>	6.1 (14.7)	1.1 (1.9)	-4.5** (2.3)	-2.7*** (0.9)	-15.6*** (5.9)
<i>Panel B: Near Violation Threshold OLS</i>					
<i>Bad Lender</i>	1.3 (19.8)	-1.9 (12.0)	6.4 (15.9)	8.3 (8.5)	3.3 (7.8)
<i>Bind</i>	5.3 (12.6)	-4.8 (5.9)	2.4 (2.3)	0.5 (0.8)	-3.1 (4.8)
<i>Bad Lender</i> $\times$ <i>Bind</i>	-13.7 (21.2)	6.3 (5.6)	-14.1** (6.7)	-3.0* (1.7)	-14.3** (6.7)
<i>Panel C: Full Sample IV</i>					
<i>Bad Lender</i>	2.2 (7.4)	-0.8 (7.9)	-1.6 (5.5)	-2.2 (3.9)	-2.0 (2.9)
<i>Bind</i>	4.3 (5.5)	1.4 (4.0)	0.7 (3.6)	0.3 (0.5)	-3.1 (9.2)
<i>Bad Lender</i> $\times$ <i>Bind</i>	5.9 (12.4)	4.2 (4.9)	-8.4** (3.7)	-4.9** (2.3)	-16.9** (7.9)
Year, Industry FE	Yes	Yes	Yes	Yes	Yes
Borrower controls	Yes	Yes	Yes	Yes	Yes
Distance control	Panel B	Panel B	Panel B	Panel B	Panel B
2008-09 supervisor control	Panel C	Panel C	Panel C	Panel C	Panel C
Controls $\times$ <i>Bad Lender</i>	Yes	Yes	Yes	Yes	Yes
Dep. var. source	SNC	Compu.	Compu.	Compu.	Compu.
Panels A&C observations	1,525	756	756	756	756
Panel B observations	357	152	152	152	152

Figure 6: Table VII: Financial and real adjustment.

5.6 Tables VIII–IX and Figure 3: robustness

Read:

- Table VIII shows the main interaction survives alternative lender-health measures, including June 2007 lead, syndicate-weighted, Lehman, ABX, and trading-revenue measures.
- Table IX shows robustness to reweighting, only best-rated loans, dropping existing violators, requiring at least two years maturity, excluding loans fully sold by the lead, and adding alternative covenant data sources.
- The baseline Table IX interaction is 22.9, and the alternative specifications generally remain around 18.8 to 30.3 percentage points.
- Figure 3 shows that dropping one lender at a time does not eliminate the effect; confidence intervals generally remain away from zero.

Economic message. The result is not an artifact of a single lender, a single lender-health proxy, or a single sample definition.

Table VIII
Robustness to Lender Health Measure

The table reports linear probability model regressions of the form $Y_{i,t,t+1} = \beta_0 + \beta_1[Bad\ Lender] + \beta_2[Bind] + \beta_3[Bad\ Lender \times Bind] + \gamma'X_{i,t,t+1} + \epsilon_{i,t,t+1}$, where *Bad Lender* is normalized to lie on the unit interval, with a value of 1 corresponding to the least healthy lender, and *Bind* is an indicator variable equal to 1 if a borrower violated a covenant in either the current or previous year. Column (1) reproduces column (3) of Table IV. In column (2), the sample and variable definitions are the same as in column (1) except that the sample excludes loans to borrowers without a loan in SNC as of June 2007 and lender health assignment is based on the lead lender as of June 2007. In columns (3) to (6), the sample and variable definitions are the same as in column (1) except that lender health assignment is based on the weighted mean health of banks in the crisis syndicate (3), the weighted mean health of the crisis syndicate imputing the health of the lead for nonbanks (4), the weighted median health of banks in the crisis syndicate (5), or the weighted median health of the crisis syndicate imputing the health of the lead for nonbanks (6). In columns (7) to (9), the sample is the same as in column (1) but lender health is defined using only the Lehman exposure measure (7), the ABX exposure (8), or the trading revenue measure (9). In column (10), the sample excludes loans that disappear by the end of the year and the dependent variable is the change in the fraction of the loan commitment from the lead lender. Reported coefficients are multiplied by 100. Borrower controls: log assets, leverage, risk rating. Loan controls: loan purpose, loan type. Standard errors two-way clustered by borrower and lead lender are reported in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Dependent Variable:	Credit Reduced									Change in Lead Share
Lender Health Based on:	Crisis Lead (Base-line)	June 2007 Lead	Crisis Syndicate-Weighted Mean	Crisis Syndicate-Weighted Median	Crisis Lead (Lehman)	Crisis Lead (ABX)	Crisis Lead (Trading)	Crisis Lead (Base-line)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>Bad Lender</i>	5.8 (12.1)	-3.6 (4.7)	-12.5 (8.3)	-5.1 (8.3)	-10.5 (6.8)	-7.1 (6.2)	0.6 (11.1)	-10.9 (12.5)	-10.9 (9.8)	7.3 (8.4)
<i>Bind</i>	5.6** (2.7)	7.6** (3.3)	-2.1 (6.5)	2.0 (4.9)	3.3 (5.2)	5.7 (4.1)	5.2** (2.4)	4.6** (2.3)	5.3** (2.4)	-4.3** (2.0)
<i>Bad Lender × Bind</i>	22.9*** (6.7)	24.9*** (6.1)	31.8*** (11.9)	26.7*** (9.6)	21.0** (9.2)	19.2** (7.8)	24.8*** (6.0)	24.2*** (5.0)	23.4*** (5.5)	-10.3** (4.8)
Impute nonbank using lead	n.a.	n.a.	No	Yes	No	Yes	n.a.	n.a.	n.a.	n.a.
Year, Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Borrower, Loan Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls × <i>Bad Lender</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,420	2,844	3,420	3,420	3,420	3,420	3,420	3,420	3,420	2,289

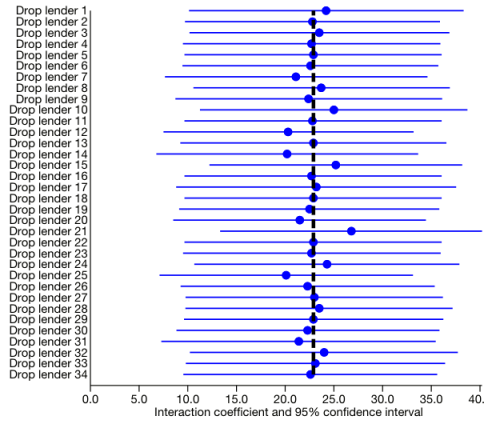
Figure 7: Table VIII: Robustness to lender health measure.

Table IX
Sample Robustness

The table reports linear probability model regressions of the form $Y_{i,t,t+1} = \beta_0 + \beta_1[Bad\ Lender] + \beta_2[Bind] + \beta_3[Bad\ Lender \times Bind] + \gamma'X_{i,t,t+1} + \epsilon_{i,t,t+1}$. Column (1) reproduces column (3) of Table IV. Column (2) reweights the sample to match the distribution of risk ratings in the full SNC universe. Column (3) only includes loans with a precrisis rating of "Best." Column (4) excludes loans that violated a covenant before the crisis. Column (5) only includes loans with at least two years maturity remaining. Column (6) excludes loans for which the lead lender does not retain any part of its share. Columns (7) and (8) expand the sample by including information on covenant compliance from Nini, Smith, and Sufi (2012) and from DealScan, respectively. The dependent variable *Credit Reduced* equals 1 if the loan is terminated before maturity or the loan commitment is reduced. *Bad Lender* is the rank of the lead lender's health normalized to lie on the unit interval, with a value of 1 corresponding to the least healthy lender. *Bind* is an indicator variable equal to 1 if a borrower violated a covenant in either the current or previous year. Reported coefficients are multiplied by 100. Borrower controls: log assets, leverage, risk rating. Loan controls: loan purpose, loan type. Standard errors two-way clustered by borrower and lead lender are reported in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Sample:	Dependent Variable: Credit Reduced							
	Baseline	Rewighted	Only "Best" Rating	Drop if Existing Violator	2+ Years Maturity	Drop if All Sold	Add NSS	Add DealScan
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Bad Lender</i>	5.8 (12.1)	7.3 (14.7)	-4.7 (9.9)	7.6 (16.5)	-2.2 (11.9)	6.4 (14.6)	2.5 (5.8)	4.6 (6.3)
<i>Bind</i>	5.6** (2.7)	4.7* (2.8)	1.1 (5.9)	1.9 (4.6)	4.2 (3.7)	1.9 (3.0)	9.3*** (2.8)	6.6** (2.8)
<i>Bad Lender × Bind</i>	22.9*** (6.7)	21.6*** (6.1)	24.4** (9.4)	29.8*** (8.5)	30.3*** (7.3)	29.7*** (8.0)	19.6*** (6.8)	18.8*** (6.6)
Year, Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Borrower, loan controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls × <i>Bad Lender</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,420	3,420	1,572	2,735	2,702	2,805	4,765	5,005

(a) Table IX: Sample robustness



(b) Figure 3: Dropping one lender at a time

5.7 Tables X–XII: placebo, heterogeneity, and aggregation

Read:

- Table X placebo tests do not show the same interaction in the precrisis period, at a size threshold, or at a shifted covenant threshold.
- Table XI shows stronger effects for credit lines and for loans with high lead share, small syndicates, and concentrated syndicates.
- These heterogeneity results are consistent with renegotiation being easier when the lead lender has more control or coordination costs are lower.
- Table XII shows the aggregate importance: the covenant channel accounts for 4.9% of total committed credit in 2008 and 5.2% in 2009.
- The corresponding bank-level decline due to lender health is 8.6% in 2008 and 6.1% in 2009.

Economic message. The channel is not only statistically robust; it is quantitatively important for aggregate credit supply during the crisis.

Table X
Placebo Specifications

The table reports regressions of the form $Y_{i,b,f,t} = \beta_0 + \beta_1[Bad_Lender] + \beta_2[Bind] + \beta_3[Bad_Lender \times Bind] + \gamma'X_{i,b,t} + \epsilon_{i,b,f,t}$. In all columns, the variable *Bad Lender* is the rank of the crisis health of the assigned lender as of the period indicated in the table header normalized to lie on the unit interval, with a value of 1 corresponding to the least healthy lender. In columns (1) and (2), the sample consists of loans in the SNC covenant sample at the start of 2006 or 2007 with at least one year maturity remaining, the dependent variable is based on outcomes in 2006 and 2007, and lender health assignment is based on the lead lender at the start of the crisis. In column (3), *Bind* is replaced with an indicator for whether the borrower is in the bottom quartile of the size distribution. In column (4), *Bind* is replaced with an indicator for whether *Distance* is below the median on each side of the covenant threshold. Reported coefficients are multiplied by 100. Borrower controls: log assets, leverage, risk rating. Loan controls: loan purpose, loan type. Standard errors two-way clustered by borrower and lead lender are reported in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Placebo:	Precrisis	Precrisis	Threshold: Size	Threshold: Shifted
Dependent Variable:	<i>Credit Reduced</i>	<i>%Δ Total Committed</i>	<i>Credit Reduced</i>	<i>Credit Reduced</i>
	(1)	(2)	(3)	(4)
<i>Bad Lender</i>	-2.2 (13.6)	2.0 (14.3)	-1.9 (5.8)	13.7* (8.0)
<i>Bind</i>	7.1** (3.0)	-16.2** (7.7)	3.3 (4.1)	-1.7 (6.9)
<i>Bad Lender × Bind</i>	8.3 (10.7)	-4.8 (17.6)	1.0 (6.6)	2.3 (14.8)
Year, Industry FE	Yes	Yes	Yes	Yes
Borrower, Loan Controls	Yes	Yes	Yes	Yes
Distance control	No	No	No	Yes
Controls × <i>Bad Lender</i>	Yes	Yes	Yes	Yes
Close violator sample	No	No	No	Yes
Observations	2,478	2,478	3,420	780

(a) Table X: Placebo specifications

Table XI
Heterogeneity

The table reports linear probability model regressions of the form $Y_{i,b,f,t} = \beta_0 + \beta_1[Bad_Lender] + \beta_2[Bind] + \beta_3[Bad_Lender \times Bind] + \gamma'X_{i,b,t} + \beta_{0,I}[I] + \beta_{1,I}[Bad_Lender \times I] + \beta_{2,I}[Bind \times I] + \beta_{3,I}[Bad_Lender \times Bind \times I] + \gamma'_I[X_{i,b,t} \times I] + \epsilon_{i,b,f,t}$. The sample is the same as Table IV. The dependent variable *Credit Reduced* equals 1 if the loan is terminated before maturity or the loan commitment is reduced. *Bad Lender* is the rank of the lead lender's health normalized to lie on the unit interval, with a value of 1 corresponding to the least healthy lender. *Bind* is an indicator variable equal to 1 if a borrower violated a covenant in either the current or previous year. In column (1), *I* is an indicator for whether the loan is a credit line. In column (2), *I* is an indicator for whether the lead lender's share of the loan commitment is above the sample median. In column (3), *I* is an indicator for whether the number of syndicate members is below the sample median. In column (4), *I* is an indicator variable for whether the Herfindahl index of loan commitment shares is above the sample median. Reported coefficients are multiplied by 100. Borrower controls: log assets, leverage, risk rating. Loan controls: loan purpose, loan type. Standard errors two-way clustered by borrower and lead lender are reported in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Interaction Variable <i>I</i> :	Dependent Variable: <i>Credit Reduced</i>			
	Credit Line	High Lead Share	Small Syndicate	Concentrated Syndicate
	(1)	(2)	(3)	(4)
<i>Bad Lender × Bind</i>	16.6** (7.7)	14.5* (8.3)	6.0 (11.4)	8.0 (9.8)
<i>Bad Lender × Bind × I</i>	16.5 (11.7)	24.7** (11.1)	28.4** (11.0)	24.1** (10.3)
Main effects	Yes	Yes	Yes	Yes
Year, Industry FE	Yes	Yes	Yes	Yes
Borrower, Loan Controls	Yes	Yes	Yes	Yes
Controls × <i>Bad Lender</i>	Yes	Yes	Yes	Yes
All controls × <i>I</i>	Yes	Yes	Yes	Yes
Observations	3,420	3,420	3,420	3,420

(b) Table XI: Heterogeneity

Table XII
Aggregate Importance

The first row reports the ratio $-(\beta_3 \sum_t Bad_Lender_b \times Bind_{t,t-1} \times Commit_{t,t-1}) / (\sum_t Commit_{t,t-1})$, where β_3 comes from the reweighted regression shown in column (2) of Table IA.VI and $Commit_{t,t-1}$ denotes the beginning-of-year commitment reweighted to make the covenant sample representative of the SNC universe along the dimension of risk rating. The second row reports the ratio $-(\gamma \sum_b Bad_Lender_b \times Commit_{b,t-1}) / (\sum_b Commit_{b,t-1})$, where γ comes from a regression of bank-level loan commitment growth on *Bad Lender*.

	<i>t</i> = 2008	<i>t</i> = 2009
	(1)	(2)
1. SNC covenant sample decline due to interaction	4.9%	5.2%
2. SNC bank-level decline due to <i>Bad Lender</i>	8.6%	6.1%

Figure 10: Table XII: Aggregate importance.

6 Short summary

- The paper studies how bank distress transmits to existing borrowers through loan covenants.
- Most bank loans had long maturities during the 2008–2009 crisis, so ordinary maturity rollover cannot explain all credit contraction.
- Loan covenants solve this puzzle because violations give lenders control rights before maturity.
- The paper uses supervisory SNC data with direct covenant compliance information.
- Covenant violations are common: roughly one-third of loans breached a covenant during the crisis.
- Lender health is measured using crisis exposure through Lehman, ABX/mortgage exposure, and trading revenue.
- Borrowers of healthier and less healthy lenders are balanced on many precrisis characteristics.
- The baseline regression interacts lender health with covenant violation status.
- The main interaction coefficient is large: unhealthy lenders are much more likely to reduce credit after a covenant violation.
- The result appears in OLS, near-threshold, and IV designs.
- The channel affects credit quantities at both loan and borrower levels.
- Borrowers do not fully substitute toward other debt or bond issuance.
- Borrowers squeezed through the covenant channel cut payouts, capital expenditures, and employment.
- Robustness checks include alternative health measures, sample restrictions, placebos, and dropping one lender at a time.
- Aggregation implies the covenant channel accounts for a majority of the cross-sectional variation in bank credit supply in the crisis years.

7 Conclusion

7.1 Core conclusion

Chodorow-Reich and Falato show that loan covenants are an important transmission mechanism from bank health to nonfinancial firms. Even when loans have long stated maturities, covenant violations give lenders the contractual power to renegotiate or reduce credit. During the financial crisis, less healthy lenders used this power more aggressively.

7.2 Economic intuition

A long-term loan is not a completely fixed supply of credit. Covenants embed state-contingent control rights. When the borrower breaches a covenant, the lender can force renegotiation. If the lender's own balance sheet is weak, the lender has a stronger incentive to shrink commitments and preserve liquidity. Therefore, the same borrower violation can lead to different outcomes depending on lender health.

7.3 Takeaway

The paper changes how we should think about bank lending transmission. Bank health can affect existing long-term credit not only through loan maturity, new lending, or refinancing, but also through covenant enforcement. The loan covenant channel turns borrower distress and lender distress into a powerful joint mechanism for credit contraction and real-economy effects.